Course Name: Electrical Circuits (2)
Instructors: Dr A. Alzyoud and Eng J. Abu-Ghalune

Date: Jan. 31. 2022
Exam Time: 90 minutes

Prob.. 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Prob. 1: Choose the correct answer.

Consider the circuit shown in Figure 1, where the voltage source has $V_S = 5\angle 0^\circ\ V_{rms}$.

1. The open circuit voltage (V_o) between the a-b terminals is
   a. $50(1-j1)$ V b. $75(1-j1)$ V c. $25(1-j1)$ V d. None of all
2. The short circuit current (I_{sc}) when the a-b terminals are shorted is
   a. -20 A b. -10 A c. -30 A d. None of all
3. The Thevenin impedance (Z_{th}) between the a-b terminals is
   a. $2.5 + j2.5\ \Omega$ b. $2.5 - j2.5\Omega$ c. $-2.5 + j2.5\ \Omega$ d. None of all
4. The current through a load of $7.5 - j2.5\ \Omega$ connected between the a – b terminals is
   a. $5(1-j1)$ A b. $10(1-j1)$ A c. $15(1-j1)$ A d. None of all

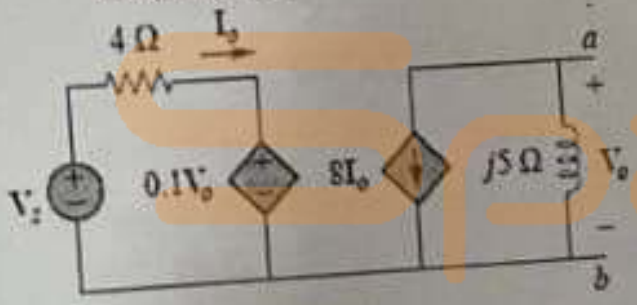


Figure 1

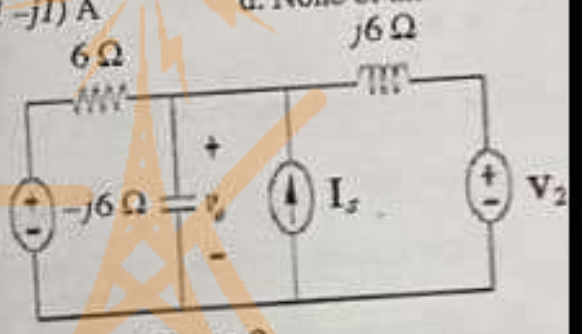


Figure 2

Consider the circuit shown in Figure 2 where $V_1 = 6\angle 0^\circ\ V_{rms}$, $V_2 = 6\angle 0^\circ\ V_{rms}$, ***and*** $I_S = 1\angle 0^\circ\ A_{rms}$

5. The voltage across the capacitor due to V_1 source only (in V) is
   a. 6 b. 3 c. 12 d. None of all
6. The voltage across the capacitor due to V_2 source only (in V) is
   a. $-j12$ b. $-j6$ c. $-j3$ d. None of all
7. The voltage across the capacitor due to I_S source only (in V) is
   a. 6 b. 12 c. 18 d. None of all

Three Loads connected in parallel across a voltage source of $200\angle 0^\circ\ V_{rms}$ ***where the loads have: Load 1: absorbs 250VA at 0.6 lagging p.f., Load 2: absorbs 200W at 0.8 leading p.f., and Load 3: ab 300VA and 150VAR.***

8. The power factor of the load 3 is
   a. 0.866 lagging b. 0.707 lagging c. 0.5 lagging d. None of all
9. The current of load 1 is
   a. $1.3 - j0.75$ A b. $0.75 - j1.0$ A c. $1 + j0.75$ A d. None of all
10. The complex power supplied by the source is
    a. $564.8 + j501.2$ VA b. $609.8 + j200$ VA c. $231.5 + j564.8$ VA d. None of all
11. The power factor of the equivalent load is
    a. 0.95 lagging b. 0.748 lagging c. 0.38 lagging d. None of all
12. The current supplied by the voltage source in rms is
    a. $2.822 - j2.506$ A b. $2.822 - j1.157$ A c. $3.049 - j1.0$ A d. None of all
13. The impedance of the equivalent load is
    a. $24.87 - j60.67\ \Omega$ b. $59.22 + j19.42\ \Omega$ c. $60.67 + j24.87\ \Omega$ d. None of all

Prob.. 1	14	15	16	17	18	19	20	21	22	23	24	25	
Answer													

A balanced Δ-load of $Z_\Delta = 8.66 + j5\ \Omega$ is supplied by a balanced Y voltages of $V_{an} = 250\angle 0^\circ\ V_{rms}$ in positive sequence.

14. The voltage V_{AB} is
a. $375 + j216.5$ V b. $216.5 + j375$V c. 433 V d. None of all

15. The phase current I_{BC} is
a. $37.5 + j21.65$ A b. $j43.3$ A c. $-21.65 - j37.5$ A d. None of all

16. The line current I_c is
a. $37.5 + j65$A b. $j75$ A c. $37.5 - j65$ A d. None of all

17. The total average power supplied by the source is
a. 48.71 KW b. 28.21 KW c. 84.37 KW d. None of all

18. The total complex power consumed by the load is
a. $56.48 + j50.12$ KVA b. $56.48 + j23.15$ KVA c. $48.71 + j28.12$ KVA d. None of all

Consider a series RLC circuit has $R = 4\ \Omega$, $L = 4mH$, $C = 0.4\mu F$, and $V_s = 12\angle 0^\circ\ V_{rms}$

19. The resonant frequency (w_o) in (rad/sec) is
a. 25 K b. 50 K c. 20 K d. None of all

20. The quality factor (Q) is
a. 50 b. 30 c. 25 d. None of all

21. The cutoff frequencies w_1 , and w_2 (in rad/sec) are
a. (19.2, 20.8) K b. (24, 26) K c. (49, 51) K d. None of all

22. The voltage across the capacitor (V_C) at resonance (in V) is
a. 400 b. 300 c. 12 d. None of all

Consider the circuit shown in Figure 3, where $V_S = 15\angle 0^\circ\ V_{rms}$, $R_1 = 2\ \Omega$, and $R_L = 4\ \Omega$.

23. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
a. $4 + j3$ b. $3 + j3$ c. $5 + j3$ d. None of all

24. The current supplied by the source (I_1)
a. $2 - j1.2$ b. $1.6 - j1.2$ c. $2.5 - j2.5$ d. None of all

25. The power absorbed by the R_L resistor is
a. 12.5 W b. 10.88 W c. 8 W d. None of all

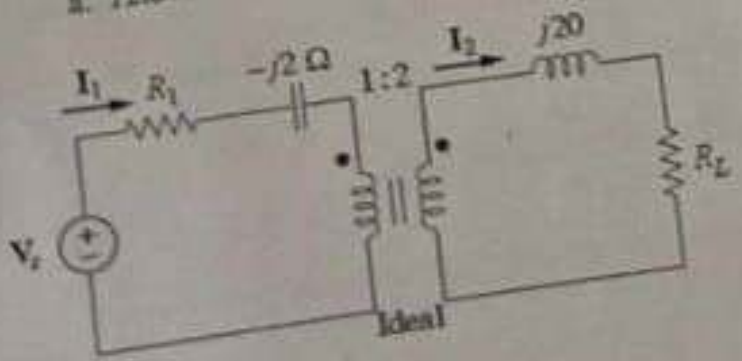


Figure 3

Prob. 1: Choose the correct answer.

1. When the impedance $Z = 8 - j6$ then the voltage across it ..... the current through it by .....
   a. leads, 36.87° b. lags , 36.87° c. leads , 53.13° d. lags , 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(wt + 26.87°)$ ***and*** $i(t) = 10\cos(wt + 10°)$.

2. The equivalent impedance at the terminals of the voltage source is
   a. $16 + j12\ \Omega$ b. $8 + j6\ \Omega$ c. $20 + j15\ \Omega$ d. $12 + j9\ \Omega$

3. When $w = 10000000000$ ==
   a. 15 mH c. 9 mH d. 6 mH

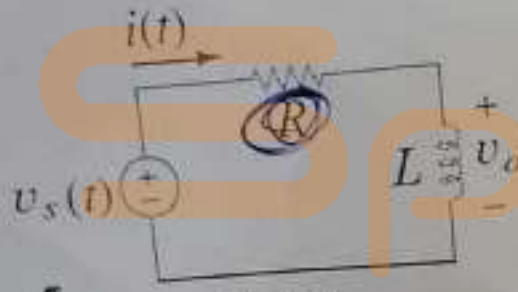


Fig. B.1

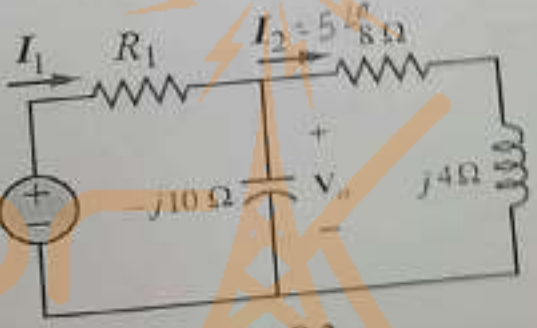


Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4\ \Omega$.

4. The equivalent impedance (Z_{eq}) seen by the source is
   a. $12 - j4\ \Omega$ b. $16 - j4\ \Omega$ c. $10 - j4\ \Omega$ d. $14 - j4\ \Omega$

5. When $I_2 = 5\angle 0°$ A then $I_1 =$
   a. $6 + j8$ A b. $3 + j4$ A c. $12 + j16$ A d. $9 + j12$ A

6. When $V_o = 10\angle 0°$ V then $V_s =$
   a. $18 + j4$ V b. $14 + j2$ V c. $16 + j3$ V d. $12 + j1$ V

SPOT POLYTECHNIC

Instructors: Dr A. Alzyoud

Prob. 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	a		c				C		e	a	b	c	d

Prob. 1: Choose the correct answer.

1. When the impedance $\mathbf{Z} = 8 - j6$ then the voltage across it ..... the current through it by .....
   a. leads, 36.87° b. lags , 36.87° c. leads , 53.13° d. lags , 53.13°

Consider the circuit shown in Fig. B.1, where $v_s(t) = 150\cos(\omega t + 26.87°)$ ***and*** $i(t) = 10\cos(\omega t - 10°)$.

2. The equivalent impedance at the terminals of the voltage source is
   a. $16 + j12\ \Omega$ b. $8 + j6\ \Omega$ c. $20 + j15\ \Omega$ d. $12 + j9\ \Omega$

3. When $\omega = 10000000000$ =
   a. 15 mH c. 9 mH d. 6 mH

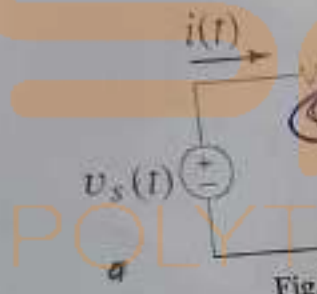


Fig. B.1



Fig. B.2

Consider the circuit shown in Fig. B.2, where $R_1 = 4\ \Omega$.

4. The equivalent impedance ($\mathbf{Z}_{eq}$) seen by the source is
   a. $12 - j4\ \Omega$ b. $16 - j4\ \Omega$ c. $10 - j4\ \Omega$ d. $14 - j4\ \Omega$

5. When $I_2 = 5\angle 0°$ A then I_1 =
   a. $6 + j8$ A b. $3 + j4$ A c. $12 + j16$ A d. $9 + j12$ A

6. When $V_o = 10\angle 0°$ V then V_s =
   a. $18 + j4$ V b. $14 + j2$ V c. $16 + j3$ V d. $12 + j1$ V

Consider the circuit shown in Fig. B.3, where $V_1 = 4\angle 0°$ V *and* $V_2 = 3\angle 0°$ V. *(Use nodal method)*

7. The voltage across the capacitor (V_x) is
   a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V

8. The current (I_1) is
   a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A

   The current (I_2) is
   a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

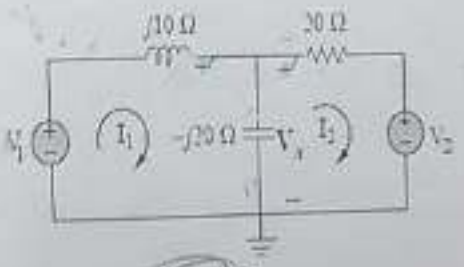


Fig. B.3

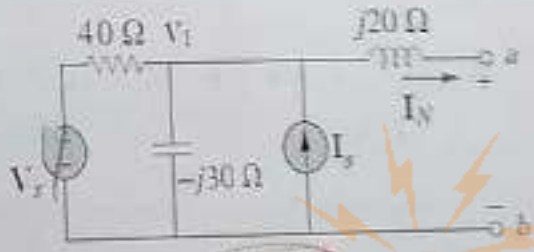


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_S = 15\angle 0°$ V *and* $I_S = 0.5\angle 90°$ A.

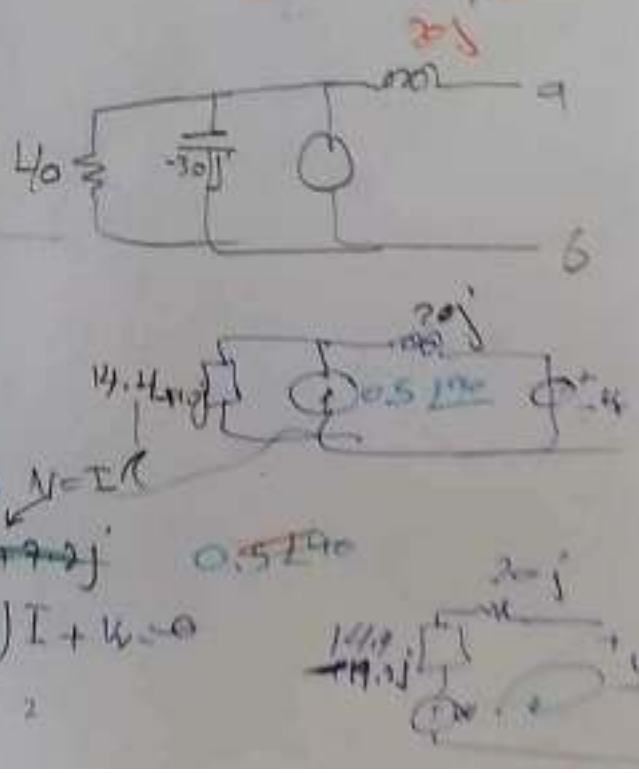
10. The voltage V_x due to the current source only
    a. $14.4 + j10.8$ V b. $4.8 + j3.6$ V c. $19.2 + j14.4$ V d. $9.6 + j7.2$ V

11. The voltage V_x due to the voltage source only
    a. $9 - j12$ V b. $7.2 - j9.6$ V c. $5.4 - j7.2$ V d. $3.6 - j4.8$ V

12. The value of V_x due to both sources
    a. 15 V b. $42.4 - j12$ V c. $8.4 - j1.2$ V d. $21.6 - j1.2$ V

13. The Thevenin equivalent impedance between the *a-b* terminals is
    a. $14.4 + j5.8\ \Omega$ b. $40 + j25\ \Omega$ c. $j4\ \Omega$ d. $14.4 - j19.2\ \Omega$

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Consider the circuit shown in Fig. B.3, where $V_1 = 4\angle 0°$ V *and* $V_2 = 3\angle 0°$ V. *(Use nodal method)*

7. The voltage across the capacitor (V_x) is
   a. $5 + j1$ V b. $5 + j3$ V c. $5 - j3$ V d. $5 - j1$ V

8. The current (I_1) is
   a. $0.3 + j0.3$ A b. $-0.3 - j0.3$ A c. $0.1 + j0.1$ A d. $-0.1 - j0.1$ A

   The current (I_2) is
   a. $0.05 - j0.15$ A b. $0.1 - j0.05$ A c. $0.15 + j0.05$ A d. $0.2 + j0.15$ A

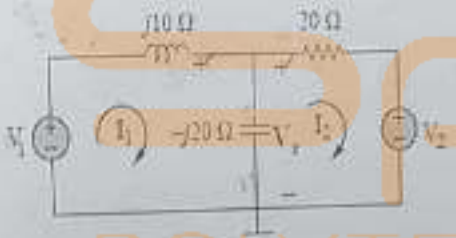


Fig. B.3

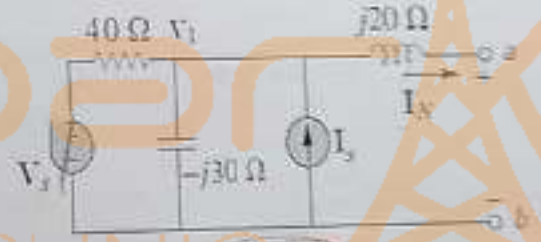


Fig. B.4

Consider the circuit shown in Fig. B.4, where $V_S = 15\angle 0°$ V *and* $I_S = 0.5\angle 90°$ A.

10. The voltage V_x due to the current source only
    a. $14.4 + j10.8$ V b. $4.8 + j3.6$ V c. $19.2 + j14.4$ V d. $9.6 + j7.2$ V

11. The voltage V_x due to the voltage source only
    a. $9 - j12$ V b. $7.2 - j9.6$ V c. $5.4 - j7.2$ V d. $3.6 - j4.8$ V

12. The value of V_x due to both sources
    a. 15 V b. $42.4 - j12$ V c. $8.4 - j1.2$ V d. $21.6 - j1.2$ V

13. The Thevenin equivalent impedance between the *a-b* terminals is
    a. $14.4 + j5.8\ \Omega$ b. $40 + j25\ \Omega$ c. $j4\ \Omega$ d. $14.4 - j19.2\ \Omega$

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Prob. 1: Consider the circuit shown in Fig. 1,

1. The load impedance (Z_L), in Ω, that absorbs the maximum average power is:

2. The open circuit voltage between *a-b* terminals (V_{oc}), when the load impedance is disconnected, is:

3. The short circuit current (I_{sc}), when *a-b* terminals are shorted is:

4. The maximum average power transferred to the load impedance (Z_L) is:

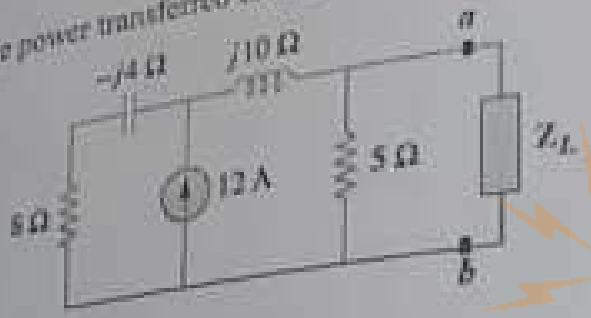


Fig.1 *(8 marks)*

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_c, is:

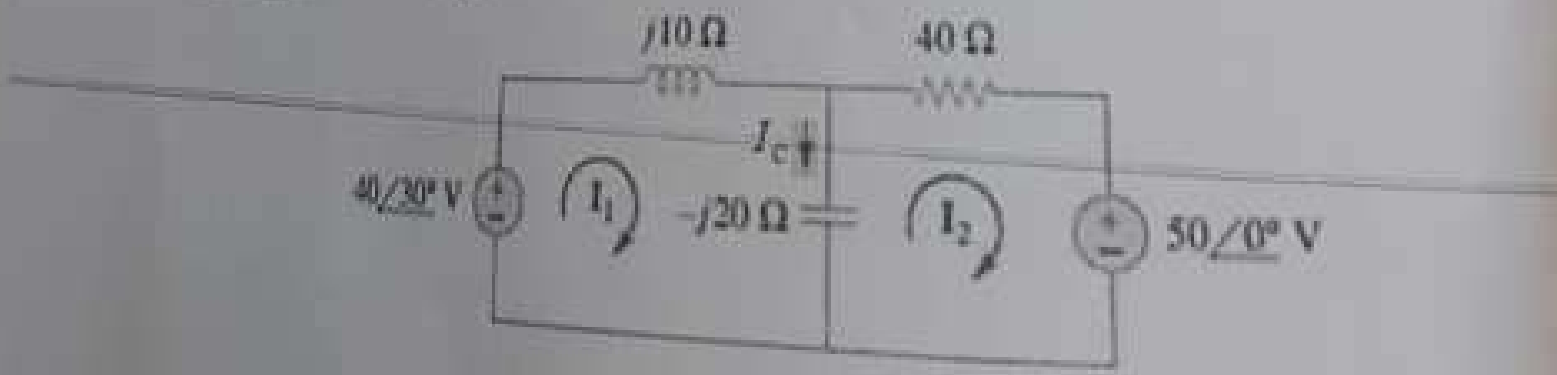


Fig 2

(8 marks)

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Prob 3.: *Two loads are connected in parallel across a voltage source of* $100\angle 0^{\circ} V_{rms}$, *where load 1 absorbs 500VAR at 0.6 lagging p.f. while load 2 absorbs 400VA at 0.75 leading p.f*

1. The impedance of load 1 (Z_1), in Ω, is:

2. The current of load 2 (I_2), in Amps, is:

3. The complex power (S) supplied by the source is:

4. The power factor of the equivalent load is:

5. The capacitance needed to raise the pf to unity is:

6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: ***Consider the circuit shown in Fig. 3,***

1. The equivalent impedance seen by the source, Z_{in}, is:

2. The current supplied by the source, I_1, is:

3. The load current, I_2, is:

4. The voltage across the primary winding, V_1, is:

5. The average power absorbed by the 2Ω resistor is:

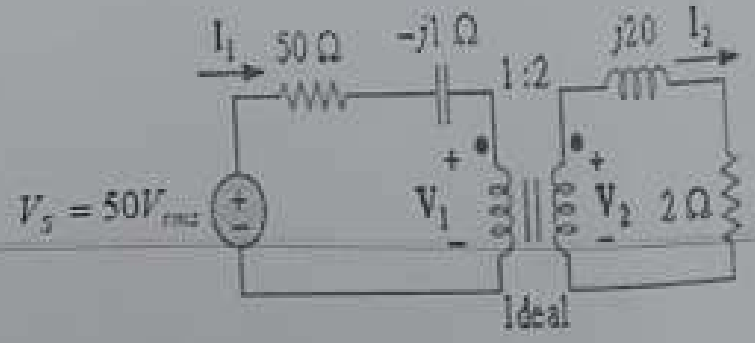


Fig.3

(10 marks)

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